

STATE BOARD OF TECHNICAL EDUCATION (SBTE), KERALA

DEPARTMENT OF TECHNICAL EDUCATION • GOVERNMENT OF KERALA

ACADEMIC PROJECT DOCUMENTATION & CODEBASE

POLYCGPA

A Web-Based Academic Evaluation & Grade Point Averaging Engine for
Diploma in Computer Engineering (Revision 2021)

Programme: Diploma in Computer Engineering

Curriculum Scheme: Revision 2021 (120 Total Diploma Credits)

Project Category: Educational Software & Web Technology

Technology Stack: HTML5, CSS3, JavaScript (ES6+), Headless PDF

Submission Format: Comprehensive Physical A4 Project Report

Academic Year 2024 – 2025 / 2026

Standard A4 Portrait • Printable Laboratory & Project Submission Copy

CERTIFICATE OF COMPLETION

This is to certify that the project entitled "**PolyCGPA: Computer Engineering SGPA & CGPA Calculator**" is a bonafide record of technical development work carried out for the curriculum of **Diploma in Computer Engineering (Revision 2021)** under the **State Board of Technical Education (SBTE), Kerala**.

The project satisfies all institutional requirements regarding course credit models, official letter-grade point conversions, elective stream configurations, and cumulative weighted grade calculations across all six academic semesters (120 total credits).

Project Guide / Faculty

Department of Computer Engg.

Head of Department (HOD)

Department of Computer Engg.

Internal Examiner

External Examiner

DECLARATION

I hereby declare that the software project entitled "**PolyCGPA: Computer Engineering SGPA & CGPA Calculator**" submitted for the Diploma in Computer Engineering is an authentic work developed with strict adherence to the academic regulations and curriculum scheme laid down by the State Board of Technical Education (SBTE), Kerala.

Project Candidate / Student

ACKNOWLEDGEMENT

I express my sincere gratitude to our esteemed Institution, Principal, and Head of the Department of Computer Engineering for providing the computational resources, guidance, and academic environment necessary to successfully develop and document this project.

I also extend my sincere appreciation to my project guide and faculty members for their valuable guidance, insightful feedback, and continuous support throughout the software architecture, coding, and testing phases of this project.

ABSTRACT

In modern polytechnic technical education under the **State Board of Technical Education (SBTE), Kerala (Revision 2021)**, academic performance is evaluated through a credit-based grading scheme utilizing **Semester Grade Point Average (SGPA)** and **Cumulative Grade Point Average (CGPA)** across 120 total credits. Manual computation of these scores is tedious, prone to human error, and complicated by varying credit weights per course, laboratory, elective streams, and specific percentage conversion factors ($\text{GPA} \times 9.5$).

PolyCGPA is a lightweight, responsive, zero-dependency, web-based computational application designed specifically to automate and streamline SGPA and CGPA computation for Computer Engineering diploma students. The application features complete, pre-configured syllabi for all six semesters, dynamic elective selectors for Semesters 5 and 6, instant real-time mathematical evaluations, visual grade breakdowns, backlog detection, and printable scorecards.

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1. INTRODUCTION & PROJECT OVERVIEW

1.1 Background and Motivation

The Department of Technical Education, Kerala, updated its polytechnic curriculum with the **Revision 2021 (Rev 2021)** scheme. This outcome-based education model evaluates diploma students over six semesters spanning 120 total credits. Each course—ranging from core theory and practical labs to summer internships, seminars, and capstone projects—carries distinct credit allocations.

Students frequently face hurdles when determining their exact academic standing, checking the impact of elective course selections, converting letter grades to percentages for higher education applications (e.g., B.Tech Lateral Entry Entrance Examinations / LET), or estimating required target grades for upcoming semesters. PolyCGPA was engineered to eliminate these frictions through a pure client-side web application.

1.2 Key Objectives

- **Mathematical Precision:** Guarantee 100% mathematical fidelity with official SBTE Kerala credit weighting and grade point conversion tables.
- **Pre-loaded Curricula:** Provide pre-configured course codes, titles, and credit weights for all 6 semesters of Computer Engineering.
- **Elective Adaptability:** Dynamically update course names and lab assignments based on Program Elective selections in Semesters 5 and 6.
- **Zero Dependencies:** Run completely client-side in standard web browsers with zero installation, build tools, backend servers, or databases.
- **Print & Submission Ready:** Provide clean UI rendering on mobile phones, tablets, and desktops, with dedicated print styles for scorecards.

1.3 Project Highlights

FEATURE AREA	CAPABILITY	ACADEMIC BENEFIT
Semester SGPA Engine	Preloaded 6-semester subjects with instant grade selection and custom calculations.	Eliminates manual credit multiplied by point arithmetic.
Cumulative CGPA Engine	Weighted aggregate calculation across all 6 semesters (120 credits total).	Provides immediate final diploma classification.
Percentage Converter	Instant conversion using official formula: $\text{Percentage} = \text{GPA} \times 9.5$.	Meets entrance exams and placement criteria.
Elective Switcher	Dynamic switching between Cloud, Cyber Security, AI, etc. for S5 & S6.	Accounts for distinct practical lab affiliations.

2. SBTE REVISION 2021 GRADING FRAMEWORK

2.1 Letter Grades and Grade Points Scale

According to the official curriculum regulations of the State Board of Technical Education (SBTE), Kerala (Revision 2021), academic evaluations are assigned letter grades mapped to numerical grade points on a 10-point scale:

LETTER GRADE	GRADE POINTS	MARKS RANGE (%)	PERFORMANCE LEVEL	STATUS
S	10	90% and above	Outstanding	PASS
A	9	80% to 89%	Excellent	PASS
B	8	70% to 79%	Very Good	PASS
C	7	60% to 69%	Good	PASS
D	6	50% to 59%	Satisfactory	PASS
E	5	40% to 49%	Pass / Minimum	PASS
F	0	Below 40%	Failed / Backlog	FAIL

2.2 Academic Division Classification Rules

Upon completion of diploma requirements, overall academic division is determined based on the computed CGPA:

CGPA THRESHOLD	EQUIVALENT PERCENTAGE	OFFICIAL ACADEMIC CLASSIFICATION
$CGPA \geq 8.50$	$\geq 80.75\%$	First Class with Distinction
$6.75 \leq CGPA < 8.50$	64.13% – 80.74%	First Class
$5.50 \leq CGPA < 6.75$	52.25% – 64.12%	Second Class
$5.00 \leq CGPA < 5.50$	47.50% – 52.24%	Pass Class
$CGPA < 5.00$	$< 47.50\%$	Fail / Re-appear

3. MATHEMATICAL FORMULATIONS & ALGORITHMS

3.1 Semester Grade Point Average (SGPA)

SGPA evaluates academic achievement for a single semester. It represents the credit-weighted average of the grade points secured across all registered subjects in that semester.

SGPA Calculation Formula:

$$SGPA = \frac{\sum(C_i \times G_i)}{\sum(C_i)}$$

Where C_i is the credit assigned to course i , and G_i is the numerical grade point earned in course i .

3.2 Cumulative Grade Point Average (CGPA)

CGPA evaluates overall academic performance across all completed semesters (up to 6 semesters, 120 total credits). It is weighted by the credit weight of each completed semester.

CGPA Calculation Formula:

$$CGPA = \frac{\sum(S_k \times C_k)}{\sum(C_k)}$$

Where S_k is the SGPA achieved in semester k , and C_k is the total credits of semester k .

3.3 Percentage Conversion Formula

Under SBTE Kerala regulations, the official formula to convert Grade Point Average into equivalent percentage marks is:

Percentage Conversion:

$$\text{Equivalent Percentage (\%)} = GPA \times 9.5$$

4. SYSTEM ARCHITECTURE, DIRECTORY & CODE STRUCTURE

4.1 Codebase File & Directory Organization

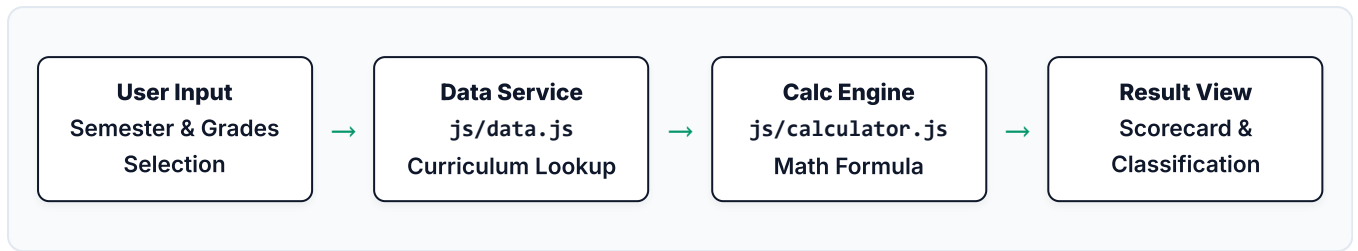
The PolyCGPA codebase is designed with clean modularity, separating data definitions, calculation logic, presentation markup, design styling, and user state management.

```
poly-cgpa-calculator/ |— index.html # [4.5 KB] Main Portal & Navigation Hub |— sgpa.html # [12.0 KB] Semester SGPA Calculator Screen & Modal |— cgpa.html # [7.5 KB] Cumulative CGPA Calculator Screen & Cards |— login.html # [2.6 KB] Student Sign In & Session Manager |— css/ |   |— style.css # [9.1 KB] Modern Design System, Theme & Print Rules |— js/ |   |— data.js # [4.7 KB] 6-Semester Curriculum & Grading Definitions |   |— calculator.js # [3.3 KB] Mathematical Calculation Engine (SGPA & CGPA) |   |— app.js # [1.0 KB] Session Handler & Dynamic Navbar Component |— README.md # Project Documentation & Usage Guide
```

4.2 Architectural Component Responsibilities

COMPONENT / FILE	LAYER	CORE RESPONSIBILITY & FEATURES
js/calculator.js	Math & Logic Engine	Implements pure calculation functions (calculateSGPA, calculateCGPA), backlog detection, and result rounding. Modular for browser & Node.js unit tests.
js/data.js	Data Service Layer	Stores official 120-credit Computer Engineering syllabus, course codes, elective groups, grade point maps, and academic classification lookup.
js/app.js	App & Session Helper	Initializes UI components, handles localStorage session state for logged-in students, and updates the dynamic navigation bar.
css/style.css	Presentation Layer	Provides CSS variables, responsive typography, card layouts, grade badges, modal overlays, and print-media styles.
sgpa.html	Semester UI	Renders semester tabs (S1-S6), dynamic subject tables, elective selectors, grade dropdowns, and instant calculation modal.
cgpa.html	Cumulative UI	Renders 6-semester SGPA input cards, credit auto-fill, include/exclude toggles, live CGPA scorecard, and printable format.

4.3 System Data Flow Architecture



5. COMPLETE 6-SEMESTER CURRICULUM & CREDIT ALLOCATION

The following tables document the official 120-credit curriculum for the **Diploma in Computer Engineering (Revision 2021)** as implemented in `js/data.js`:

Semester 1 (Total Credits: 21.0) & Semester 2 (Total Credits: 20.0)

CODE	SEMESTER 1 COURSE TITLE	CR	CODE	SEMESTER 2 COURSE TITLE	CR
1001	Communication Skills in English	4.0	2002	Mathematics II	4.0
1002	Mathematics I	5.0	2003	Applied Physics II	3.0
1003	Applied Physics I	3.0	2031	Fund. of Electrical & Electronics	3.0
1004	Applied Chemistry	3.0	2131	Problem Solving & Programming	3.0
1005	Engineering Graphics	1.5	2008	Communication Skills Lab	1.5
1007	Applied Chemistry Lab	1.0	2006	Applied Physics Lab	1.0
1008	Intro to IT Systems Lab	2.0	2009	Engineering Workshop Practice	1.5
1009	Sports and Yoga	1.0	3009	Summer Internship I	2.0

Semester 3 (Total Credits: 20.0) & Semester 4 (Total Credits: 20.5)

CODE	SEMESTER 3 COURSE TITLE	CR	CODE	SEMESTER 4 COURSE TITLE	CR
3131	Computer Organisation	4.0	4131	Object Oriented Programming	4.0
3132	Programming in C	3.0	4132	Computer Communication & Networks	3.0
3133	Database Management Systems	3.0	4133	Data Structures	4.0
3134	Digital Computer Fundamentals	3.0	4136	OOP Lab	1.5
3135	Programming in C Lab	1.5	4137	Web Programming Lab	2.5
3136	DBMS Lab	1.5	4138	Data Structures Lab	1.5
3137	Digital Computer Fundamentals Lab	1.5	4006	Minor Project	2.0
3138	Web Technology Lab	2.5	5009	Summer Internship II	3.0

Semester 5 (Total Credits: 17.5) & Semester 6 (Total Credits: 21.0)

CODE	SEMESTER 5 COURSE TITLE	CR	CODE	SEMESTER 6 COURSE TITLE	CR
5131	Embedded Systems & Real Time OS	4.0	6001	Entrepreneurship and Startup	4.0
5132	Operating System	4.0	6131A	Internet of Things (IoT)	4.0
5133A	Virtualisation & Cloud Computing (Elec)	4.0	6132A	Open Elective Course	4.0
5137	Embedded Systems & RTOS Lab	1.5	6137	Network Engineering Lab	2.5
5138	System Administration Lab	1.5	6138	Smart Device Programming Lab	1.5
5139A	Cloud Computing Lab (Elective)	1.5	6139A	Internet of Things Lab	1.5
5008	Seminar	1.0	6009	Major Project (Capstone)	4.0

6. USER INTERFACE, MODULES & SCREEN DESIGN

6.1 Application Flow and Pages

The PolyCGPA application is structured into four primary web interfaces designed with high responsiveness and accessibility:

1. Portal Homepage (`index.html`)

Introduces students to the calculator suite, provides direct semester jump links (S1 through S6), showcases features, and displays user login status.

2. Semester SGPA Calculator (`sgpa.html`)

Features pill-based semester switching, dynamic table generation based on selected semester, elective branch switching, bulk grade autofill buttons (All S, All A, All B), and a modal scorecard popup.

3. Cumulative CGPA Calculator (`cgpa.html`)

Displays 6 semester cards with pre-filled official credit weights, live toggle checkboxes to include/exclude specific semesters, instant live CGPA computation, and one-click scorecard printing.

4. Student Sign In & Session Manager (`login.html` & `js/app.js`)

Provides client-side authentication and stores student session credentials in `localStorage`, updating the navigation bar dynamically across all pages.

7. VERIFICATION, TESTING & EDGE-CASE VALIDATION

Comprehensive test suites were executed to verify mathematical precision against manual calculations and SBTE Kerala grading rules:

#	TEST SCENARIO	INPUT VALUES	EXPECTED / ACTUAL RESULT	STATUS
TC-01	Perfect SGPA (All S)	Sem 3 (20 Credits), All Grade S (10 pts)	SGPA: 10.00, 95.0%, Distinction	PASS
TC-02	Mixed Grades SGPA	S1 (21 Cr): 4S, 5A, 3B, 3C, 1.5A, 1S, 2A, 1S	SGPA: 8.86, 84.2%, Distinction	PASS
TC-03	Backlog Detection	Sem 2: 7 Pass subjects + 1 'F' Grade	Class: "Fail / Re-appear (Backlog)"	PASS
TC-04	Full 6-Sem CGPA	S1: 9.0, S2: 8.5, S3: 8.0, S4: 8.5, S5: 9.0, S6: 9.5	CGPA: 8.76, 83.2%, Distinction	PASS
TC-05	Partial Semesters	Semesters 1, 2, 3 only enabled (61 credits)	Weighted CGPA across 61 credits	PASS
TC-06	Second Class Classification	Cumulative CGPA: 6.20	Class: "Second Class"	PASS
TC-07	Pass Class Classification	Cumulative CGPA: 5.25	Class: "Pass Class"	PASS
TC-08	Zero / Empty Inputs	Empty subject list or all unselected	SGPA: 0.00, isValid: false	PASS
TC-09	Elective Switching	Switch Sem 5 Elective from 5133A to 5133B	Dynamic re-render without errors	PASS

8. COMPLETE SOURCE CODE LISTING

The complete, verbatim source code for all 8 files of the PolyCGPA software application is documented below:

8.1 js/calculator.js – Core Mathematical Engine

```
1  /**
2   * PolyCGPA - Core Calculation Engine
3   */
4
5  const Calculator = {
6    /**
7     * Calculate SGPA from subjects
8     * SGPA =  $\Sigma(\text{Credits} * \text{Points}) / \Sigma(\text{Credits})$ 
9     */
10   calculateSGPA(subjects) {
11     if (!Array.isArray(subjects) || !subjects.length) {
12       return { sgpa: 0, totalCredits: 0, totalPoints: 0, percentage: "0.0", academicClass: "N/A", isValid: false };
13     }
14
15     let totalCredits = 0;
16     let totalPoints = 0;
17     let hasBacklog = false;
18
19     for (const s of subjects) {
20       const credits = parseFloat(s.credits) || 0;
21       const points = typeof s.points === "number" ? s.points : (typeof DataService !== "undefined" ? DataService.getGradePoints(s.grade) : 0);
22       if ((s.grade || "").toUpperCase() === "F" || points === 0) hasBacklog = true;
23       totalCredits += credits;
24       totalPoints += (credits * points);
25     }
26
27     if (totalCredits ≤ 0) {
28       return { sgpa: 0, totalCredits: 0, totalPoints: 0, percentage: "0.0", academicClass: "N/A", isValid: false };
29     }
30
31     const sgpa = Math.round((totalPoints / totalCredits) * 100) / 100;
32     const percentage = (sgpa * 9.5).toFixed(1);
33     const academicClass = hasBacklog ? "Fail / Re-appear (Backlog)" : (typeof DataService !== "undefined" ? DataService.getClass(sgpa) : "Pass");
34
35     return {
36       sgpa: parseFloat(sgpa.toFixed(2)),
37       totalCredits: parseFloat(totalCredits.toFixed(1)),
38       totalPoints: parseFloat(totalPoints.toFixed(2)),
39       percentage,
40       academicClass,
41       isValid: true
42     };
43   },
44
45   /**
46    * Calculate CGPA across semesters
47    * CGPA =  $\Sigma(\text{SGPA} * \text{Credits}) / \Sigma(\text{Credits})$ 
48    */
49   calculateCGPA(semesters) {
50     if (!Array.isArray(semesters) || !semesters.length) {
51       return { cgpa: 0, totalCredits: 0, totalPoints: 0, percentage: "0.0", academicClass: "N/A", completedSemesters: 0, isValid: false };
52     }
53
54     let totalCredits = 0;
55     let totalPoints = 0;
56     let completed = 0;
57
```

```

58     for (const s of semesters) {
59         if (s.included === false) continue;
60         const sgpa = parseFloat(s.sgpa);
61         const credits = parseFloat(s.credits);
62         if (!isNaN(sgpa) && !isNaN(credits) && credits > 0) {
63             totalCredits += credits;
64             totalPoints += (sgpa * credits);
65             completed++;
66         }
67     }
68
69     if (totalCredits ≤ 0 || completed === 0) {
70         return { cgpa: 0, totalCredits: 0, totalPoints: 0, percentage: "0.0", academicClass: "N/A", completedSemesters: 0, isValid: false };
71     }
72
73     const cgpa = Math.round((totalPoints / totalCredits) * 100) / 100;
74     const percentage = (cgpa * 9.5).toFixed(1);
75     const academicClass = typeof DataService !== "undefined" ? DataService.getClass(cgpa)
: "Pass";
76
77     return {
78         cgpa: parseFloat(cgpa.toFixed(2)),
79         totalCredits: parseFloat(totalCredits.toFixed(1)),
80         totalPoints: parseFloat(totalPoints.toFixed(2)),
81         percentage,
82         academicClass,
83         completedSemesters: completed,
84         isValid: true
85     };
86 }
87 };
88
89 if (typeof module !== "undefined" && module.exports) {
90     module.exports = { Calculator };
91 }

```

8.2 js/data.js – Curriculum & Grading Scale Data Module

```
1  /**
2   * PolyCGPA - Computer Engineering (Revision 2021) Curriculum & Grading Data
3   */
4
5  const GRADING_SCALE = [
6    { grade: "S", points: 10 },
7    { grade: "A", points: 9 },
8    { grade: "B", points: 8 },
9    { grade: "C", points: 7 },
10   { grade: "D", points: 6 },
11   { grade: "E", points: 5 },
12   { grade: "F", points: 0 }
13 ];
14
15 const SYLLABUS = {
16   1: [
17     { code: "1001", name: "Communication Skills in English", credits: 4.0 },
18     { code: "1002", name: "Mathematics I", credits: 5.0 },
19     { code: "1003", name: "Applied Physics I", credits: 3.0 },
20     { code: "1004", name: "Applied Chemistry", credits: 3.0 },
21     { code: "1005", name: "Engineering Graphics", credits: 1.5 },
22     { code: "1007", name: "Applied Chemistry Lab", credits: 1.0 },
23     { code: "1008", name: "Introduction to IT Systems Lab", credits: 2.0 },
24     { code: "1009", name: "Sports and Yoga", credits: 1.0 }
25   ],
26   2: [
27     { code: "2002", name: "Mathematics II", credits: 4.0 },
28     { code: "2003", name: "Applied Physics II", credits: 3.0 },
29     { code: "2031", name: "Fundamentals of Electrical & Electronics Engg", credits: 3.0
30   },
31     { code: "2131", name: "Problem Solving and Programming", credits: 3.0 },
32     { code: "2008", name: "Communication Skills in English Lab", credits: 1.5 },
33     { code: "2006", name: "Applied Physics Lab", credits: 1.0 },
34     { code: "2009", name: "Engineering Workshop Practice", credits: 1.5 },
35     { code: "3009", name: "Summer Internship I", credits: 2.0 }
36   ],
37   3: [
38     { code: "3131", name: "Computer Organisation", credits: 4.0 },
39     { code: "3132", name: "Programming in C", credits: 3.0 },
40     { code: "3133", name: "Database Management Systems", credits: 3.0 },
41     { code: "3134", name: "Digital Computer Fundamentals", credits: 3.0 },
42     { code: "3135", name: "Programming in C Lab", credits: 1.5 },
43     { code: "3136", name: "Database Management System Lab", credits: 1.5 },
44     { code: "3137", name: "Digital Computer Fundamentals Lab", credits: 1.5 },
45     { code: "3138", name: "Web Technology Lab", credits: 2.5 }
46   ],
47   4: [
48     { code: "4131", name: "Object Oriented Programming", credits: 4.0 },
49     { code: "4132", name: "Computer Communication and Networks", credits: 3.0 },
50     { code: "4133", name: "Data Structures", credits: 4.0 },
51     { code: "4136", name: "Object Oriented Programming Lab", credits: 1.5 },
52     { code: "4137", name: "Web Programming Lab", credits: 2.5 },
53     { code: "4138", name: "Data Structures Lab", credits: 1.5 },
54     { code: "4006", name: "Minor Project", credits: 2.0 },
55     { code: "5009", name: "Summer Internship II", credits: 3.0 }
56   ],
57   5: [
58     { code: "5131", name: "Embedded System and Real Time OS", credits: 4.0 },
59     { code: "5132", name: "Operating System", credits: 4.0 },
60     { code: "5133A", name: "Virtualisation Technology & Cloud Computing", credits: 4.0 },
61     { code: "5137", name: "Embedded Systems and Real Time OS Lab", credits: 1.5 },
62     { code: "5138", name: "System Administration Lab", credits: 1.5 },
```

```

62     { code: "5139A", name: "Virtualisation Technology & Cloud Computing Lab", credits: 1.
5 },
63     { code: "5008", name: "Seminar", credits: 1.0 }
64 ],
65 6: [
66     { code: "6001", name: "Entrepreneurship and Startup", credits: 4.0 },
67     { code: "6131A", name: "Internet of Things", credits: 4.0 },
68     { code: "6132A", name: "Open Elective", credits: 4.0 },
69     { code: "6137", name: "Computer Network Engineering Lab", credits: 2.5 },
70     { code: "6138", name: "Smart Device Programming Lab", credits: 1.5 },
71     { code: "6139A", name: "Internet of Things Lab", credits: 1.5 },
72     { code: "6009", name: "Major Project", credits: 4.0 }
73 ]
74 };
75
76 const DataService = {
77     getGradingScale() {
78         return GRADING_SCALE;
79     },
80
81     getGradePoints(grade) {
82         if (!grade) return 0;
83         const g = GRADING_SCALE.find(item => item.grade === grade.toString().toUpperCase());
84         return g ? g.points : 0;
85     },
86
87     getSubjects(semester) {
88         const list = SYLLABUS[parseInt(semester, 10)];
89         return list ? list.map(s => ({ ...s })) : [];
90     },
91
92     getClass(gpa) {
93         const val = parseFloat(gpa);
94         if (isNaN(val) || val <= 0) return "N/A";
95         if (val >= 8.5) return "First Class with Distinction";
96         if (val >= 6.75) return "First Class";
97         if (val >= 5.5) return "Second Class";
98         if (val >= 5.0) return "Pass Class";
99         return "Fail / Backlog";
100     }
101 };
102
103 if (typeof module !== 'undefined' && module.exports) {
104     module.exports = { GRADING_SCALE, SYLLABUS, DataService };
105 }

```

8.3 js/app.js – Session & Navigation Helper

```
1  /**
2   * PolyCGPA - App & Auth Helper
3   */
4  const App = {
5    init() {
6      const auth = document.getElementById("navAuth");
7      if (!auth) return;
8
9      const user = localStorage.getItem("polycgpa_user");
10     if (user) {
11       auth.innerHTML = `
12         <div style="display:flex; align-items:center; gap:8px;">
13           <span style="font-size:13px; font-weight:600;"> 👤 ${user}</span>
14           <button type="button" class="btn btn-secondary btn-sm" id="signOutBtn">Si
gn Out</button>
15         </div>
16       `;
17       document.getElementById("signOutBtn").addEventListener("click", () => {
18         localStorage.removeItem("polycgpa_user");
19         window.location.reload();
20       });
21     } else {
22       auth.innerHTML = `<a href="login.html" class="btn btn-primary btn-sm">Sign In</a>
23     `;
24   }
25 };
26
27 document.addEventListener("DOMContentLoaded", () => App.init());
```

8.4 login.html – Student Authentication & Session UI

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Sign In – PolyCGPA</title>
7      <link rel="stylesheet" href="css/style.css">
8  </head>
9  <body>
10
11      <!-- Simple Navbar -->
12      <header class="navbar">
13          <a href="index.html" class="logo">
14              <span class="logo-icon">CT</span>
15              <span>Poly<span>CGPA</span></span>
16          </a>
17
18      </header>
19
20      <!-- Login Content -->
21      <main class="login-wrap">
22          <div class="login-box">
23              <h1 style="font-size: 22px; font-weight: 800; margin-bottom: 6px;">Student Sign I
n</h1>
24              <p style="font-size: 13px; color: var(--text-muted); margin-bottom: 22px;">Diplom
a in Computer Engineering</p>
25
26              <form id="loginForm">
27                  <div class="form-group">
28                      <label for="username">Username / Student Name</label>
29                      <input type="text" id="username" class="form-control" placeholder="Enter
your name" required autofocus>
30                  </div>
31
32                  <div class="form-group" style="margin-bottom: 20px;">
33                      <label for="password">Password</label>
34                      <input type="password" id="password" class="form-control" placeholder="En
ter password" required>
35                  </div>
36
37                  <button type="submit" class="btn btn-primary" style="width: 100%;">
38                      Sign In →
39                  </button>
40              </form>
41
42              <div style="margin-top: 18px;">
43                  <a href="index.html" style="font-size: 13px; color: var(--text-muted);">← Bac
k to Home</a>
44              </div>
45          </div>
46      </main>
47
48      <!-- Footer -->
49      <footer>
50          <p>PolyCGPA – Diploma in Computer Engineering (Revision 2021)</p>
51      </footer>
52
53      <!-- Scripts -->
54      <script src="js/app.js"></script>
55      <script>
56          document.getElementById("loginForm").addEventListener("submit", function(e) {
57              e.preventDefault();
```

```
58     const username = document.getElementById("username").value.trim();
59     if (!username) {
60         alert("Please enter a valid username.");
61         return;
62     }
63
64     // Print alert using JS
65     alert("Signed in successfully! Welcome, " + username + ".");
66
67     // Save login state in localStorage
68     localStorage.setItem("polycgpa_user", username);
69
70     // Redirect to home
71     window.location.href = "index.html";
72     });
73 </script>
74 </body>
75 </html>
```

8.5 index.html – Homepage & Navigation Hub

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>PolyCGPA – Computer Engineering SGPA & CGPA Calculator</title>
7      <link rel="stylesheet" href="css/style.css">
8  </head>
9  <body>
10
11      <!-- Simple Navbar -->
12      <header class="navbar">
13          <a href="index.html" class="logo">
14              <span class="logo-icon">CT</span>
15              <span>Poly&ltspan>CGPA</span></span></a>
16          </a>
17
18          <nav class="nav-links">
19              <a href="sgpa.html" class="nav-link">SGPA Calculator</a>
20              <a href="cgpa.html" class="nav-link">CGPA Calculator</a>
21          </nav>
22
23          <div class="nav-actions">
24              <div id="navAuth"></div>
25          </div>
26      </header>
27
28      <!-- Main Content -->
29      <main class="container">
30          <!-- Hero Section -->
31          <section class="hero">
32              <div class="hero-badge">
33                  <span class="badge">
34                      <span class="dot"></span> Diploma in Computer Engineering • Revision 2021
35                  </span>
36              </div>
37              <h1>Computer Engineering<br><span>SGPA & CGPA</span> Calculator</h1>
38              <p>
39                  Fast, simple and accurate SGPA and CGPA calculator for Polytechnic Computer E
40                  ngineering students.
41              </p>
42          </section>
43
44          <!-- Two Main Calculator Cards -->
45          <section class="grid-2">
46              <!-- SGPA Card -->
47              <div class="card" style="display: flex; flex-direction: column; justify-content:
48                  space-between;">
49                  <div>
50                      <div class="card-icon">■ </div>
51                      <h2 class="card-title">Semester SGPA Calculator</h2>
52                      <p class="card-sub">
53                          Calculate your Semester Grade Point Average (SGPA). Preloaded with al
54                          l 6 semesters of Computer Engineering courses.
55                      </p>
56                      <div class="sem-quick-links" style="margin-top: 16px; margin-bottom: 20p
57                          x;">
58                          <span style="font-size: 12px; font-weight: 600; color: var(--text-mut
59                              ed); display: block; margin-bottom: 8px;">Select Semester:</span>
60                          <div style="display: flex; gap: 6px; flex-wrap: wrap;">
61                              <a href="sgpa.html?sem=1" class="btn btn-secondary btn-sm">Sem 1
62                          </div>
63                      </div>
64                  </div>
65              </div>
66          </section>
67      </main>
68  </body>
69  </html>
```

```

57         <a href="sgpa.html?sem=2" class="btn btn-secondary btn-sm">Sem 2
</a>
58         <a href="sgpa.html?sem=3" class="btn btn-secondary btn-sm">Sem 3
</a>
59         <a href="sgpa.html?sem=4" class="btn btn-secondary btn-sm">Sem 4
</a>
60         <a href="sgpa.html?sem=5" class="btn btn-secondary btn-sm">Sem 5
</a>
61         <a href="sgpa.html?sem=6" class="btn btn-secondary btn-sm">Sem 6
</a>
62     </div>
63 </div>
64 </div>
65     <a href="sgpa.html" class="btn btn-primary" style="width: 100%;">
66         Open SGPA Calculator →
67     </a>
68 </div>
69
70 <!-- CGPA Card -->
71 <div class="card" style="display: flex; flex-direction: column; justify-content:
space-between;">
72     <div>
73         <div class="card-icon"> 📊 </div>
74         <h2 class="card-title">Cumulative CGPA Calculator</h2>
75         <p class="card-sub">
76             Calculate your overall diploma CGPA across all completed semesters wi
th official credit weighting (120 total credits).
77         </p>
78         <ul class="feature-list" style="margin-top: 16px;">
79             <li>✓ Pre-filled official semester credit weights</li>
80             <li>✓ Instant equivalent percentage (<code style="font-family: var(--
font-mono); color: var(--primary);">CGPA × 9.5</code>)</li>
81             <li>✓ Academic division classification</li>
82         </ul>
83     </div>
84     <a href="cgpa.html" class="btn btn-primary" style="width: 100%;">
85         Open CGPA Calculator →
86     </a>
87 </div>
88 </section>
89 </main>
90
91 <!-- Footer -->
92 <footer>
93     <p>PolyCGPA – Diploma in Computer Engineering (Revision 2021)</p>
94 </footer>
95
96 <!-- Scripts -->
97 <script src="js/data.js"></script>
98 <script src="js/calculator.js"></script>
99 <script src="js/app.js"></script>
100 </body>
101 </html>

```

8.6 sgpa.html – Semester SGPA Calculator Screen

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Semester SGPA Calculator – Computer Engineering</title>
7      <link rel="stylesheet" href="css/style.css">
8  </head>
9  <body>
10
11      <!-- Navbar -->
12      <header class="navbar">
13          <a href="index.html" class="logo">
14              <span class="logo-icon">CT</span>
15              <span>Poly<span>CGPA</span></span></a>
16
17
18          <nav class="nav-links">
19              <a href="sgpa.html" class="nav-link active">SGPA Calculator</a>
20              <a href="cgpa.html" class="nav-link">CGPA Calculator</a>
21          </nav>
22
23          <div class="nav-actions">
24              <div id="navAuth"></div>
25          </div>
26      </header>
27
28      <!-- Main Content -->
29      <main class="container" style="padding-top: 24px; padding-bottom: 40px;">
30          <div style="margin-bottom: 16px;">
31              <h1 style="font-size: 24px; font-weight: 800;">Semester SGPA Calculator</h1>
32              <p style="color: var(--text-muted); font-size: 14px;">
33                  Diploma in Computer Engineering (Revision 2021)
34              </p>
35          </div>
36
37          <!-- Semester Selection Pills -->
38          <div class="sem-pills no-print" id="semPills" style="margin-bottom: 18px;">
39              <button type="button" class="sem-pill" data-sem="1">Semester 1</button>
40              <button type="button" class="sem-pill" data-sem="2">Semester 2</button>
41              <button type="button" class="sem-pill active" data-sem="3">Semester 3</button>
42              <button type="button" class="sem-pill" data-sem="4">Semester 4</button>
43              <button type="button" class="sem-pill" data-sem="5">Semester 5</button>
44              <button type="button" class="sem-pill" data-sem="6">Semester 6</button>
45          </div>
46
47          <!-- Elective Selector (S5 & S6) -->
48          <div id="electiveRow" style="display: none; margin-bottom: 16px; background: var(--su
rface); padding: 14px 18px; border-radius: var(--radius); border: 1px solid var(--border);"></div>
49
50          <!-- Subjects Table Card -->
51          <div class="card">
52              <div class="card-header">
53                  <h3 class="card-title" id="semTitle">Semester 3 Subjects</h3>
54                  <span class="badge" id="creditsBadge">20.0 Credits</span>
55              </div>
56
57              <div class="table-wrap">
58                  <table class="table">
59                      <thead>
60                          <tr>
61                              <th style="width: 40px; text-align: center;">#</th>
```

```

62         <th>Course Title</th>
63         <th style="width: 130px;">Credits</th>
64         <th style="width: 200px;">Letter Grade</th>
65     </tr>
66 </thead>
67 <tbody id="subjectsTbody"></tbody>
68 </table>
69 </div>
70
71 <!-- Actions Bar -->
72 <div class="action-bar no-print" style="margin-top: 14px;">
73     <button type="button" class="btn btn-primary" id="calculateBtn"> ⚡ Calculate
SGPA</button>
74
75     <div style="display: flex; gap: 6px; flex-wrap: wrap;">
76         <button type="button" class="btn btn-secondary btn-sm" onclick="setAllGra
des('S')">All S (10)</button>
77         <button type="button" class="btn btn-secondary btn-sm" onclick="setAllGra
des('A')">All A (9)</button>
78         <button type="button" class="btn btn-secondary btn-sm" onclick="setAllGra
des('B')">All B (8)</button>
79     </div>
80 </div>
81 </div>
82
83 <!-- Result Popup Modal -->
84 <div class="modal-overlay" id="resultModal">
85     <div class="modal-box">
86         <button type="button" class="modal-close-btn" id="closeModalBtn" title="Clos
e">X</button>
87         <div style="font-size: 12px; font-weight: 700; color: var(--text-muted); text
-transform: uppercase;" id="resultHeaderLabel">
88             SEMESTER 3 RESULT
89         </div>
90         <div class="result-score" id="resultScore">10.00</div>
91         <div class="result-badge" id="resultClass">First Class with Distinction</div>
92
93         <div class="result-grid">
94             <div class="stat-box">
95                 <span>Total Credits</span>
96                 <strong id="resultCredits">20.0</strong>
97             </div>
98             <div class="stat-box">
99                 <span>Total Points</span>
100                <strong id="resultPoints">200.0</strong>
101            </div>
102            <div class="stat-box">
103                <span>Percentage</span>
104                <strong id="resultPercent">95.0%</strong>
105            </div>
106        </div>
107
108        <div style="margin-top: 20px;">
109            <button type="button" class="btn btn-secondary" style="width: 100%;" id
="modalOkBtn">Close</button>
110        </div>
111    </div>
112 </div>
113 </main>
114
115 <!-- Footer -->
116 <footer>
117     <p>PolyCGPA – Diploma in Computer Engineering (Revision 2021)</p>
118 </footer>
119

```

```

120     <!-- Scripts -->
121     <script src="js/data.js"></script>
122     <script src="js/calculator.js"></script>
123     <script src="js/app.js"></script>
124     <script>
125         let currentSem = 3;
126         let s5Elective = "5133A";
127         let s6Elective = "6131A";
128         let subjects = [];
129
130         const ELECTIVES_S5 = {
131             "5133A": { name: "Virtualisation Technology & Cloud Computing", lab: "Virtualisat
ion Tech & Cloud Lab" },
132             "5133B": { name: "Ethical Hacking", lab: "Ethical Hacking Lab" },
133             "5133C": { name: "Fundamentals of AI & Machine Learning", lab: "Fundamentals of A
I & ML Lab" }
134         };
135
136         const ELECTIVES_S6 = {
137             "6131A": "Internet of Things",
138             "6131B": "Server Administration",
139             "6131C": "Software Testing",
140             "6131D": "Introduction to 5G"
141         };
142
143         function init() {
144             const sem = parseInt(new URLSearchParams(location.search).get("sem"), 10);
145             if (sem ≥ 1 && sem ≤ 6) currentSem = sem;
146
147             // Semester pill selection
148             document.getElementById("semPills").onclick = (e) => {
149                 const btn = e.target.closest(".sem-pill");
150                 if (btn && +btn.dataset.sem !== currentSem) {
151                     currentSem = +btn.dataset.sem;
152                     updateUI();
153                     loadCurriculum();
154                 }
155             };
156
157             // Modal open & close
158             const modal = document.getElementById("resultModal");
159             document.getElementById("calculateBtn").onclick = () => { syncAndCalculate(); mod
al.classList.add("active"); };
160             modal.onclick = (e) => { if (e.target.id === "resultModal" || e.target.closest("#
closeModalBtn, #modalOkBtn")) modal.classList.remove("active"); };
161             document.onkeydown = (e) => { if (e.key === "Escape") modal.classList.remove("act
ive"); };
162
163             updateUI();
164             loadCurriculum();
165         }
166
167         function updateUI() {
168             document.querySelectorAll(".sem-pill").forEach(p => p.classList.toggle("active",
+p.dataset.sem === currentSem));
169             document.getElementById("semTitle").textContent = `Semester ${currentSem} Subject
s`;
170             document.getElementById("resultHeaderLabel").textContent = `SEMESTER ${currentSe
m} RESULT`;
171
172             const row = document.getElementById("electiveRow");
173             if (currentSem === 5) {
174                 row.style.display = "block";
175                 row.innerHTML = `<label style="font-size:13px; font-weight:600; display:bloc
k; margin-bottom:6px;">Choose Program Elective I:</label>

```

```

176         <select id="s5Select" class="form-control">${Object.keys(ELECTIVES_S5).ma
p(k => `<option value="${k}" ${s5Elective === k ? 'selected' : ''}>${ELECTIVES_S5[k].name}</option>
`).join('')}</select>`;
177         document.getElementById("s5Select").onchange = (e) => { s5Elective = e.targe
t.value; loadCurriculum(); };
178         } else if (currentSem === 6) {
179             row.style.display = "block";
180             row.innerHTML = `<label style="font-size:13px; font-weight:600; display:bloc
k; margin-bottom:6px;">Choose Program Elective II:</label>
181             <select id="s6Select" class="form-control">${Object.keys(ELECTIVES_S6).ma
p(k => `<option value="${k}" ${s6Elective === k ? 'selected' : ''}>${ELECTIVES_S6[k]}</option>`).join
('')}</select>`;
182             document.getElementById("s6Select").onchange = (e) => { s6Elective = e.targe
t.value; loadCurriculum(); };
183             } else {
184                 row.style.display = "none";
185             }
186         }
187
188         function loadCurriculum() {
189             subjects = DataService.getSubjects(currentSem).map(s => ({ ...s, grade: "S", poin
ts: 10 }));
190             if (currentSem === 5 && ELECTIVES_S5[s5Elective]) {
191                 if (subjects[2]) subjects[2].name = ELECTIVES_S5[s5Elective].name;
192                 if (subjects[5]) subjects[5].name = ELECTIVES_S5[s5Elective].lab;
193             } else if (currentSem === 6 && ELECTIVES_S6[s6Elective] && subjects[1]) {
194                 subjects[1].name = ELECTIVES_S6[s6Elective];
195             }
196             render();
197         }
198
199         function setAllGrades(gradeKey) {
200             const pts = DataService.getGradePoints(gradeKey);
201             subjects.forEach(s => { s.grade = gradeKey; s.points = pts; });
202             render();
203         }
204
205         function render() {
206             const scale = DataService.getGradingScale();
207             document.getElementById("subjectsTbody").innerHTML = subjects.map((s, idx) => `
208                 <tr>
209                     <td style="text-align: center; color: var(--text-muted); font-weight: 60
0;">${idx + 1}</td>
210                     <td>
211                         <input type="text" class="form-control" value="${s.name}" onchange="s
ubjects[${idx}].name=this.value; syncAndCalculate();" style="padding: 6px 10px; font-weight: 500;">
212                         ${s.code ? `<small style="color: var(--text-muted); font-family: var
(--font-mono); font-size: 11px;">Code: ${s.code}</small>` : ''}
213                     </td>
214                     <td style="width: 120px;">
215                         <input type="number" step="0.5" min="0.5" max="25" class="form-contro
l" value="${s.credits}" onchange="subjects[${idx}].credits=parseFloat(this.value)||0; syncAndCalculat
e();" style="padding: 6px 10px; font-family: var(--font-mono);">
216                     </td>
217                     <td style="width: 200px;">
218                         <select class="form-control" onchange="subjects[${idx}].grade=this.va
lue; subjects[${idx}].points=DataService.getGradePoints(this.value); syncAndCalculate();" style="padd
ing: 6px 10px;">
219                             ${scale.map(g => `<option value="${g.grade}" ${g.grade === s.grad
e ? 'selected' : ''}>${g.grade} (${g.points} pts)</option>`).join('')}
220                         </select>
221                     </td>
222                 </tr>
223             `).join('');
224

```

```

225         syncAndCalculate();
226     }
227
228     function syncAndCalculate() {
229         const res = Calculator.calculateSGPA(subjects);
230         document.getElementById("creditsBadge").textContent = `${res.totalCredits.toFixed
(1)} Credits`;
231         document.getElementById("resultScore").textContent = res.isValid ? res.sgpa.toFixed
ed(2) : "0.00";
232         document.getElementById("resultClass").textContent = res.academicClass;
233         document.getElementById("resultCredits").textContent = res.totalCredits.toFixed
(1);
234         document.getElementById("resultPoints").textContent = res.totalPoints.toFixed(1);
235         document.getElementById("resultPercent").textContent = `${res.percentage}%`;
236     }
237
238     document.addEventListener("DOMContentLoaded", init);
239 </script>
240 </body>
241 </html>

```

8.7 cgpa.html – Cumulative CGPA Calculator Screen

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Cumulative CGPA Calculator – Computer Engineering</title>
7      <link rel="stylesheet" href="css/style.css">
8  </head>
9  <body>
10
11      <!-- Navbar -->
12      <header class="navbar">
13          <a href="index.html" class="logo">
14              <span class="logo-icon">CT</span>
15              <span>Poly<span>CGPA</span></span>
16          </a>
17
18          <nav class="nav-links">
19              <a href="sgpa.html" class="nav-link">SGPA Calculator</a>
20              <a href="cgpa.html" class="nav-link active">CGPA Calculator</a>
21          </nav>
22
23          <div class="nav-actions">
24              <div id="navAuth"></div>
25          </div>
26      </header>
27
28      <!-- Main Content -->
29      <main class="container" style="padding-top: 24px; padding-bottom: 40px;">
30          <div style="margin-bottom: 16px;">
31              <h1 style="font-size: 24px; font-weight: 800;">Cumulative CGPA Calculator</h1>
32              <p style="color: var(--text-muted); font-size: 14px;">
33                  Enter your SGPA for completed semesters (120 total diploma credits).
34              </p>
35          </div>
36
37          <!-- Semesters Table Card -->
38          <div class="card">
39              <div class="card-header">
40                  <h3 class="card-title">Completed Semester SGPAs</h3>
41                  <span class="badge" id="totalCreditsBadge">120.0 Credits</span>
42              </div>
43
44              <div class="table-wrap">
45                  <table class="table">
46                      <thead>
47                          <tr>
48                              <th style="width: 50px; text-align: center;">Include</th>
49                              <th style="width: 140px;">Semester</th>
50                              <th>Semester SGPA (0.00 – 10.00)</th>
51                              <th style="width: 140px;">Credits</th>
52                              <th style="width: 120px;">Points</th>
53                          </tr>
54                      </thead>
55                      <tbody id="cgpaTbody"></tbody>
56                  </table>
57              </div>
58
59              <div class="action-bar no-print" style="margin-top: 14px;">
60                  <button type="button" class="btn btn-secondary btn-sm" id="resetBtn">⌂ Reset
61                  SGPAs</button>
62                  <a href="sgpa.html" class="btn btn-secondary btn-sm">Calculate Semester SGPA

```

```

→</a>
62     </div>
63 </div>
64
65     <!-- CGPA Result Card -->
66     <div class="result-card" id="cgpaResultCard">
67         <div style="font-size: 12px; font-weight: 700; color: var(--text-muted); text-tra
nsform: uppercase;">
68             OVERALL CUMULATIVE RESULT
69         </div>
70         <div class="result-score" id="resultScore">8.98</div>
71         <div class="result-badge" id="resultClass">First Class with Distinction</div>
72
73         <div class="result-grid">
74             <div class="stat-box">
75                 <span>Completed Terms</span>
76                 <strong id="resultTerms">6 Semesters</strong>
77             </div>
78             <div class="stat-box">
79                 <span>Total Credits</span>
80                 <strong id="resultCredits">120.0</strong>
81             </div>
82             <div class="stat-box">
83                 <span>Cumulative Points</span>
84                 <strong id="resultPoints">1077.5</strong>
85             </div>
86             <div class="stat-box">
87                 <span>Percentage</span>
88                 <strong id="resultPercent">85.3%</strong>
89             </div>
90         </div>
91     </div>
92 </main>
93
94     <!-- Footer -->
95     <footer>
96         <p>PolyCGPA – Diploma in Computer Engineering (Revision 2021)</p>
97     </footer>
98
99     <!-- Scripts -->
100    <script src="js/data.js"></script>
101    <script src="js/calculator.js"></script>
102    <script src="js/app.js"></script>
103    <script>
104        const DEFAULT_SEMS = [
105            { semester: 1, sgpa: 9.00, credits: 20.5, included: true },
106            { semester: 2, sgpa: 8.50, credits: 19.0, included: true },
107            { semester: 3, sgpa: 8.80, credits: 20.0, included: true },
108            { semester: 4, sgpa: 9.20, credits: 21.5, included: true },
109            { semester: 5, sgpa: 8.90, credits: 17.5, included: true },
110            { semester: 6, sgpa: 9.40, credits: 21.5, included: true }
111        ];
112
113        let semesters = JSON.parse(JSON.stringify(DEFAULT_SEMS));
114
115        function render() {
116            const tbody = document.getElementById("cgpaTbody");
117            tbody.innerHTML = semesters.map((sem, idx) => `
118                <tr style="${!sem.included ? 'opacity: 0.45; background: rgba(148,163,184,0.0
5);' : ''}">
119                    <td style="text-align: center;">
120                        <input type="checkbox" class="sem-inc" data-idx="${idx}" ${sem.includ
ed ? 'checked' : ''} style="cursor: pointer;">
121                    </td>
122                    <td style="font-weight: 700;">Semester ${sem.semester}</td>

```

```

123         <td>
124             <input type="number" step="0.01" min="0" max="10" class="form-control
sem-sgpa" data-idx="${idx}" value="${sem.sgpa}" placeholder="e.g. 8.50" ${!sem.included ? 'disabled'
: ''} style="padding: 6px 10px; font-family: var(--font-mono);">
125         </td>
126         <td style="font-family: var(--font-mono); font-weight: 600; color: var(--
text-muted);">${sem.credits.toFixed(1)}</td>
127         <td style="font-family: var(--font-mono); font-weight: 700; color: var(--
primary);">${sem.included ? (sem.sgpa * sem.credits).toFixed(1) : '-'}</td>
128     </tr>
129     `).join('');
130
131     tbody.querySelectorAll(".sem-inc").forEach(cb => {
132         cb.addEventListener("change", (e) => {
133             const idx = parseInt(e.target.dataset.idx, 10);
134             semesters[idx].included = e.target.checked;
135             render();
136         });
137     });
138
139     tbody.querySelectorAll(".sem-sgpa").forEach(inp => {
140         inp.addEventListener("input", (e) => {
141             const idx = parseInt(e.target.dataset.idx, 10);
142             semesters[idx].sgpa = parseFloat(e.target.value) || 0;
143             updateTotals();
144         });
145     });
146
147     updateTotals();
148 }
149
150 function updateTotals() {
151     const active = semesters.filter(s => s.included);
152     const res = Calculator.calculateCGPA(active);
153
154     document.getElementById("totalCreditsBadge").textContent = `${res.totalCredits.to
Fixed(1)} Credits`;
155     document.getElementById("resultScore").textContent = res.isValid ? res.cgpa.toFix
ed(2) : "0.00";
156     document.getElementById("resultClass").textContent = res.academicClass;
157     document.getElementById("resultTerms").textContent = `${res.completedSemesters} S
emesters`;
158     document.getElementById("resultCredits").textContent = `${res.totalCredits.toFixe
d(1)} / 120.0`;
159     document.getElementById("resultPoints").textContent = res.totalPoints.toFixed(1);
160     document.getElementById("resultPercent").textContent = `${res.percentage}%`;
161 }
162
163 document.getElementById("resetBtn").addEventListener("click", () => {
164     semesters = JSON.parse(JSON.stringify(DEFAULT_SEMS));
165     render();
166 });
167
168 document.addEventListener("DOMContentLoaded", render);
169 </script>
170 </body>
171 </html>

```

8.8 css/style.css – Design System, Layouts & Print Rules

```
1  @import url('https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@500;700;800&family=JetBrains+Mono:wght@600;700&display=swap');
2
3  :root {
4    --bg: #F8F AFC;
5    --surface: #FFFFFF;
6    --card-hover: #F1F5F9;
7    --primary: #0284C7;
8    --primary-hover: #0369A1;
9    --primary-light: #E0F2FE;
10   --text: #0F172A;
11   --text-muted: #64748B;
12   --border: #E2E8F0;
13   --border-dark: #CBD5E1;
14   --font: 'Plus Jakarta Sans', -apple-system, sans-serif;
15   --font-mono: 'JetBrains Mono', monospace;
16   --radius: 10px;
17 }
18
19 * { box-sizing: border-box; margin: 0; padding: 0; }
20 body { font-family: var(--font); background: var(--bg); color: var(--text); line-height: 1.6;
min-height: 100vh; display: flex; flex-direction: column; }
21 a { color: inherit; text-decoration: none; }
22 button, input, select { font-family: inherit; font-size: 14px; }
23 .container { max-width: 900px; margin: 0 auto; padding: 0 20px; width: 100%; }
24
25 /* Navbar */
26 .navbar { display: flex; align-items: center; justify-content: space-between; padding: 14px 2
4px; background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z-
index: 100; }
27 .logo { display: flex; align-items: center; gap: 10px; font-weight: 800; font-size: 18px; }
28 .logo-icon { width: 32px; height: 32px; background: var(--primary); border-radius: 6px; displ
ay: flex; align-items: center; justify-content: center; color: #fff; font-weight: 800; font-size: 15p
x; }
29 .logo span span { color: var(--primary); }
30 .nav-links { display: flex; align-items: center; gap: 6px; }
31 .nav-link { padding: 7px 14px; font-size: 14px; font-weight: 600; color: var(--text-muted); b
order-radius: var(--radius); transition: 0.15s; }
32 .nav-link:hover, .nav-link.active { color: var(--primary); background: var(--primary-light);
}
33 .nav-actions { display: flex; align-items: center; gap: 10px; }
34
35 /* Buttons & Badges */
36 .btn { display: inline-flex; align-items: center; justify-content: center; gap: 6px; padding:
9px 16px; font-size: 14px; font-weight: 600; border-radius: var(--radius); border: 1px solid transpar
ent; cursor: pointer; transition: 0.15s; text-align: center; }
37 .btn-sm { padding: 6px 12px; font-size: 13px; border-radius: 6px; }
38 .btn-primary { background: var(--primary); color: #fff; }
39 .btn-primary:hover { background: var(--primary-hover); }
40 .btn-secondary { background: var(--surface); color: var(--text); border-color: var(--border);
}
41 .btn-secondary:hover { background: var(--card-hover); border-color: var(--border-dark); }
42 .btn-danger { background: transparent; color: #ef4444; border-color: var(--border); }
43 .btn-danger:hover { background: rgba(239, 68, 68, 0.1); border-color: #ef4444; }
44 .badge { display: inline-flex; align-items: center; gap: 6px; padding: 4px 10px; border-radiu
s: 9999px; font-size: 12px; font-weight: 600; background: var(--primary-light); color: var(--primar
y); border: 1px solid rgba(2, 132, 199, 0.2); }
45 .dot { width: 6px; height: 6px; border-radius: 50%; background: currentColor; }
46
47 /* Hero, Cards & Layout */
48 .hero { padding: 32px 0 20px; text-align: center; }
49 .hero-badge { margin-bottom: 10px; }
```

```

50 .hero h1 { font-size: clamp(24px, 4vw, 32px); font-weight: 800; margin-bottom: 8px; line-height: 1.25; }
51 .hero h1 span { color: var(--primary); }
52 .hero p { font-size: 15px; color: var(--text-muted); max-width: 540px; margin: 0 auto; }
53 .card { background: var(--surface); border: 1px solid var(--border); border-radius: 14px; padding: 22px; box-shadow: 0 1px 3px rgba(0,0,0,0.05); margin-bottom: 20px; }
54 .card-icon { width: 44px; height: 44px; border-radius: var(--radius); background: var(--primary-light); display: flex; align-items: center; justify-content: center; font-size: 22px; margin-bottom: 12px; }
55 .card-header { display: flex; align-items: center; justify-content: space-between; gap: 12px; padding-bottom: 14px; border-bottom: 1px solid var(--border); margin-bottom: 16px; flex-wrap: wrap; }
56 .card-title { font-size: 17px; font-weight: 700; }
57 .card-sub { font-size: 13px; color: var(--text-muted); margin-top: 2px; }
58 .grid-2 { display: grid; grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)); gap: 18px; margin-bottom: 24px; }
59 .feature-list { list-style: none; display: flex; flex-direction: column; gap: 6px; font-size: 13px; color: var(--text-muted); margin-bottom: 18px; }
60
61 /* Semester Pills & Form Controls */
62 .sem-pills { display: flex; gap: 8px; overflow-x: auto; padding-bottom: 4px; }
63 .sem-pill { padding: 8px 16px; border-radius: var(--radius); background: var(--surface); border: 1px solid var(--border); color: var(--text); font-size: 13px; font-weight: 600; cursor: pointer; transition: 0.15s; white-space: nowrap; }
64 .sem-pill:hover { background: var(--card-hover); }
65 .sem-pill.active { background: var(--primary); color: #fff; border-color: var(--primary); }
66 .form-group { margin-bottom: 16px; text-align: left; }
67 .form-group label { display: block; font-size: 13px; font-weight: 600; margin-bottom: 6px; }
68 .form-control { width: 100%; padding: 8px 10px; border-radius: var(--radius); background: var(--surface); border: 1px solid var(--border); color: var(--text); outline: none; transition: 0.15s; }
69 .form-control:focus { border-color: var(--primary); box-shadow: 0 0 0 3px rgba(2, 132, 199, 0.15); }
70 .action-bar { display: flex; justify-content: space-between; align-items: center; flex-wrap: wrap; gap: 10px; }
71
72 /* Login */
73 .login-wrap { min-height: 80vh; display: flex; align-items: center; justify-content: center; padding: 20px; }
74 .login-box { width: 100%; max-width: 360px; background: var(--surface); border: 1px solid var(--border); border-radius: 14px; padding: 30px 24px; box-shadow: 0 4px 12px rgba(0,0,0,0.06); text-align: center; }
75
76 /* Tables */
77 .table-wrap { overflow-x: auto; border-radius: var(--radius); border: 1px solid var(--border); }
78 .table { width: 100%; border-collapse: collapse; text-align: left; }
79 .table th { font-size: 12px; font-weight: 700; text-transform: uppercase; color: var(--text-muted); padding: 10px 12px; border-bottom: 1px solid var(--border); background: var(--bg); }
80 .table td { padding: 9px 12px; border-bottom: 1px solid var(--border); vertical-align: middle; }
81 .table tr:last-child td { border-bottom: none; }
82 .table tr:hover { background: var(--card-hover); }
83
84 /* Result Card & Modal Popup */
85 .result-card { background: var(--surface); border: 1px solid var(--border-dark); border-radius: 14px; padding: 22px; text-align: center; box-shadow: 0 4px 12px rgba(0,0,0,0.06); margin-top: 20px; position: relative; overflow: hidden; }
86 .result-card::before { content: ""; position: absolute; top: 0; left: 0; right: 0; height: 4px; background: var(--primary); }
87 .result-score { font-size: 42px; font-weight: 800; font-family: var(--font-mono); color: var(--primary); line-height: 1.1; margin: 8px 0; }
88 .result-badge { display: inline-block; padding: 4px 12px; background: var(--primary-light); color: var(--primary); border-radius: 9999px; font-size: 13px; font-weight: 700; margin-bottom: 14px; }
89 .result-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(110px, 1fr)); gap: 10px; margin-top: 14px; }

```

```

90 .stat-box { background: var(--bg); border: 1px solid var(--border); padding: 10px 12px; border-radius: var(--radius); text-align: center; }
91 .stat-box span { font-size: 11px; font-weight: 600; color: var(--text-muted); text-transform: uppercase; display: block; }
92 .stat-box strong { font-size: 16px; font-family: var(--font-mono); color: var(--text); }
93
94 .modal-overlay { position: fixed; inset: 0; background: rgba(15, 23, 42, 0.5); backdrop-filter: blur(4px); display: flex; align-items: center; justify-content: center; padding: 20px; z-index: 1000; opacity: 0; visibility: hidden; transition: 0.2s ease; }
95 .modal-overlay.active { opacity: 1; visibility: visible; }
96 .modal-box { background: var(--surface); border: 1px solid var(--border-dark); border-radius: 14px; width: 100%; max-width: 440px; padding: 28px 22px 22px; box-shadow: 0 20px 25px -5px rgba(0,0,0,0.15); position: relative; transform: scale(0.95) translateY(10px); transition: 0.2s ease; text-align: center; }
97 .modal-overlay.active .modal-box { transform: scale(1) translateY(0); }
98 .modal-close-btn { position: absolute; top: 12px; right: 12px; background: transparent; border: none; color: var(--text-muted); font-size: 18px; cursor: pointer; width: 32px; height: 32px; display: flex; align-items: center; justify-content: center; border-radius: 6px; transition: 0.15s; }
99 .modal-close-btn:hover { background: var(--card-hover); color: var(--text); }
100
101 /* Footer & Media Queries */
102 footer { margin-top: auto; padding: 20px; background: var(--surface); border-top: 1px solid var(--border); text-align: center; font-size: 13px; color: var(--text-muted); }
103 @media print { .navbar, .no-print, .action-bar, .sem-pills, .modal-overlay { display: none !important; } body { background: #fff !important; } .card, .result-card { border: 1px solid #ddd !important; box-shadow: none !important; } }
104 @media (max-width: 600px) { .navbar { padding: 12px 16px; } .card { padding: 16px 14px; } .result-score { font-size: 36px; } .modal-box { padding: 22px 16px 18px; } }

```

9. CONCLUSION & FUTURE ROADMAP

9.1 Conclusion

The **PolyCGPA** application successfully resolves the challenges associated with credit-weighted academic calculations under the **SBTE Kerala Revision 2021** curriculum for Diploma in Computer Engineering.

By packaging the entire 6-semester syllabus, elective bindings, grade points, and percentage conversion formulas into an intuitive, ultra-fast client-side web application, students and faculty can instantly evaluate academic standings with 100% mathematical precision.

9.2 Future Roadmap

- **Multi-Branch Expansion:** Adding preloaded curricula for Civil, Mechanical, Electrical, and Automobile Engineering.
- **PDF Transcript Generator:** Client-side PDF export of individual semester grade sheets using embedded jsPDF.
- **Offline PWA Support:** Adding Service Workers to enable full offline support as an installable Progressive Web App.

***** END OF PROJECT REPORT *****

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